



Mahidol University
Wisdom of the Land

Pharmacology of Lipid-lowering Agents

Surasak Wichaiyo (Ph.D.)

Department of Pharmacology
Faculty of Pharmacy
Mahidol University

Pharmacology I (PYCH304)

20th Nov 2025 (8.00-10.00)



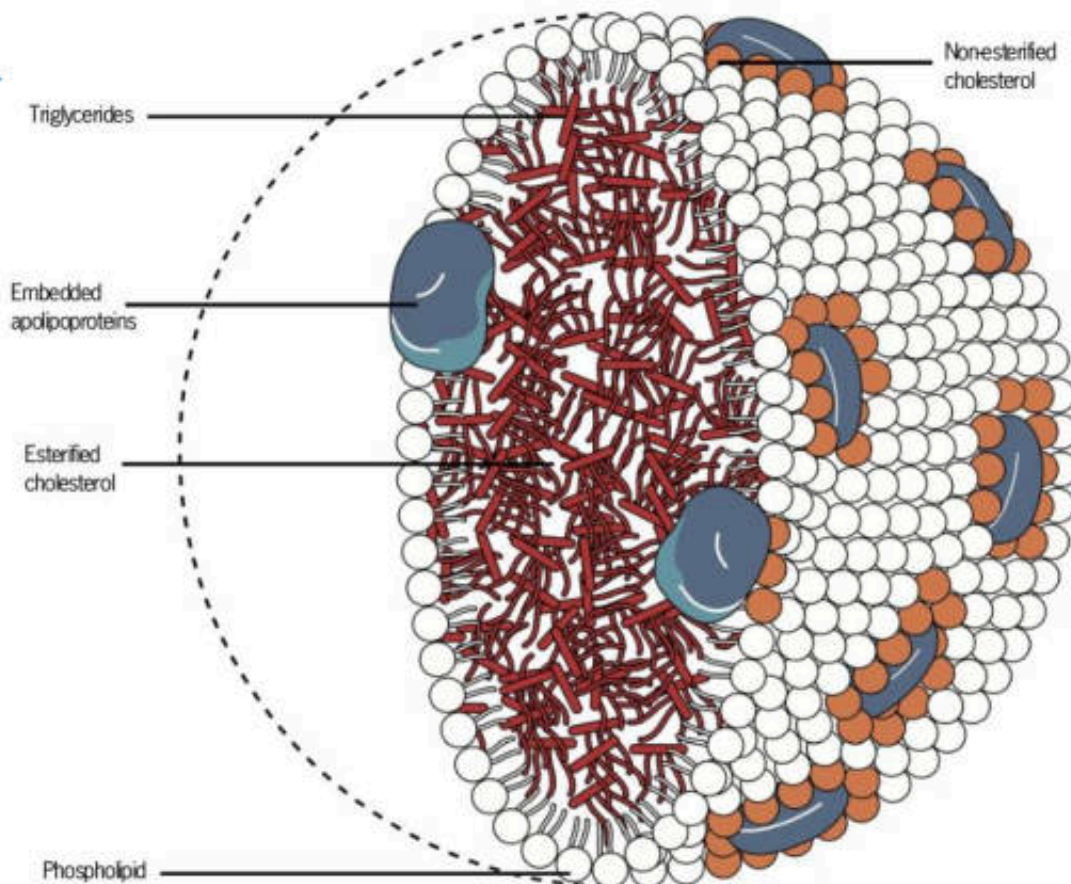
Learning Objectives

After the class, students should be able to

1. Explain the types and metabolic pathways of plasma lipid
2. Explain the mechanism of action (MOA), pharmacokinetic and pharmacodynamic properties of lipid-lowering agents
3. Explain ADRs and drug interactions of lipid-lowering agents

The components of a lipoprotein

- **Lipoproteins** = macromolecular assemblies that contain lipids and proteins
- The **lipid** constituents include
 - Free and esterified cholesterol
 - Triglycerides
 - Phospholipids
- The **protein** components
 - Known as *apolipoproteins* or *apoproteins*
 - Provide structural stability to the lipoproteins
 - May function as ligands in lipoprotein-receptor interactions or as cofactors in enzymatic processes that regulate lipoprotein



Surface

- Free cholesterol (non-esterified)
- Phospholipid (e.g. lecithin)
- Apolipoproteins (non-exchangeable and exchangeable)
- Fat-soluble antioxidants and vitamins

Core

- Cholesterol esters (fatty acyl cholesterol)
- Triglyceride (triacylglycerols)
- Fat-soluble antioxidants and vitamins

<https://bjcardio.co.uk/2015/07/lipid-s-module-1-lipid-metabolism-and-its-role-in-atherosclerosis-3/>



Characteristics of plasma lipoproteins

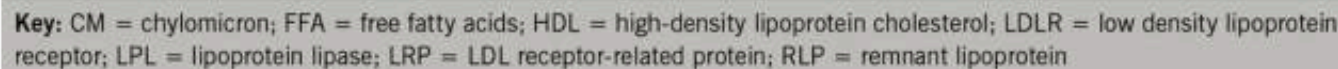
LIPOPROTEIN CLASS	DENSITY (g/mL)	MAJOR LIPID CONSTITUENT	TG:CHOL	SIGNIFICANT APOPROTEINS	SITE OF SYNTHESIS	CATABOLIC PATHWAY
Chylomicrons and remnants	<1.006	Dietary triglycerides and cholesterol	10:1	B-48, E, A-I, A-IV, C-I, C-II, C-III	Intestine	Triglyceride hydrolysis by LPL; apo E-mediated remnant uptake by liver
VLDL	<1.006	"Endogenous" or hepatic triglycerides	5:1	B-100, E, C-I, C-II, C-III	Liver	Triglyceride hydrolysis by LPL
IDL	1.006–1.019	Cholesteryl esters and "endogenous" triglycerides	1:1	B-100, E, C-II, C-III	Product of VLDL catabolism	50% converted to LDL mediated by HL; 50% apo E-mediated uptake by liver
LDL	1.019–1.063	Cholesteryl esters	NS	B-100	Product of VLDL catabolism	Apo B-100-mediated uptake by LDL receptor (~75% in liver)
HDL	1.063–1.21	Phospholipids, cholesteryl esters	NS	A-I, A-II, E, C-I, C-II, C-III	Intestine, liver, plasma	Complex: transfer of cholesteryl ester to VLDL and LDL; uptake of HDL cholesterol by hepatocytes
Lp(a)	1.05–1.09	Cholesteryl esters	NS	B-100 apo(a)	Liver	Unknown

CHOL, cholesterol; NS, not significant (triglyceride is < 5% of LDL and HDL); TG, triglyceride.



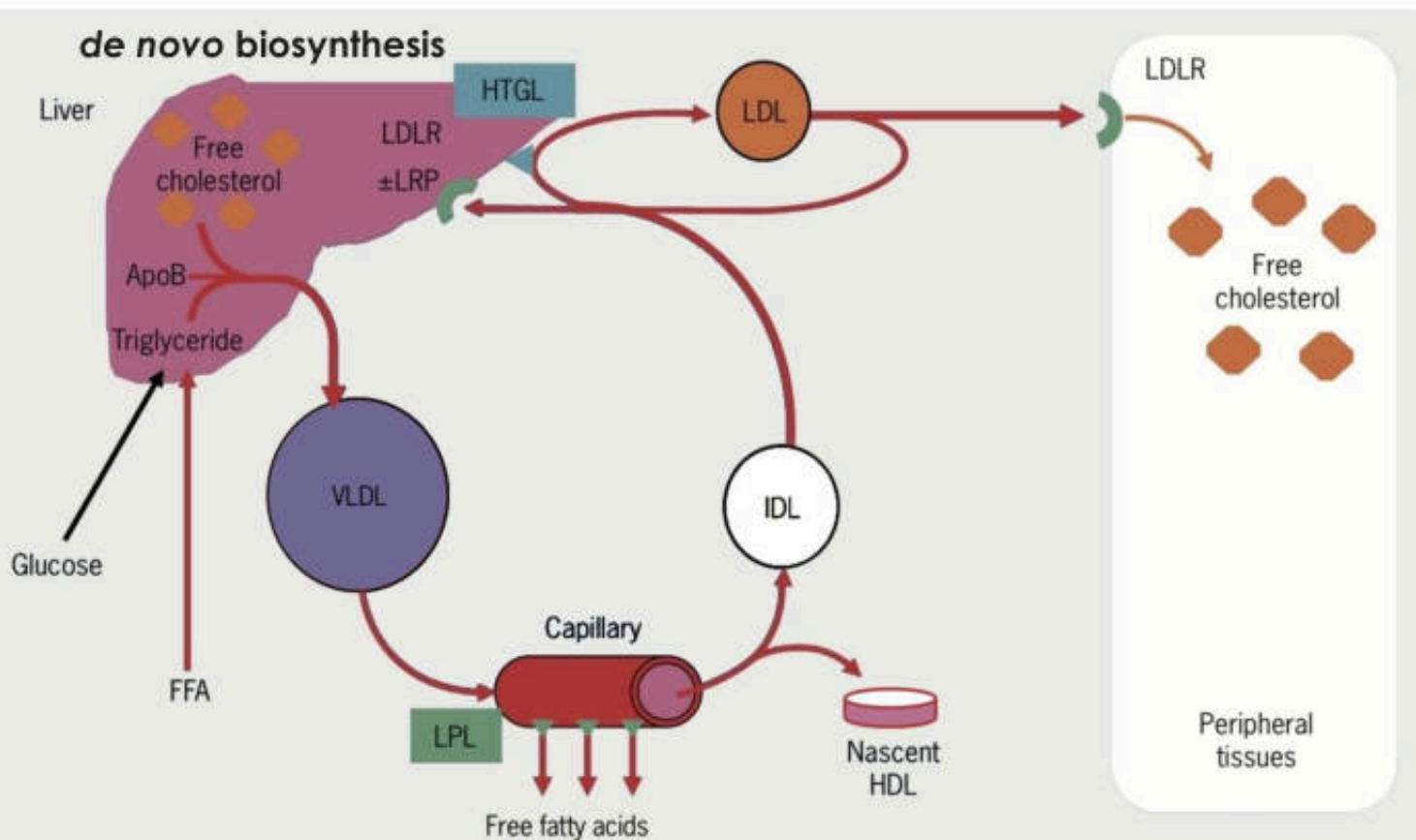
Plasma lipid pathways

- The three major pathways are coordinated by the liver to ensure that balance (homeostasis) of the major lipid classes is maintained
 - Exogenous
 - Endogenous
 - Reverse





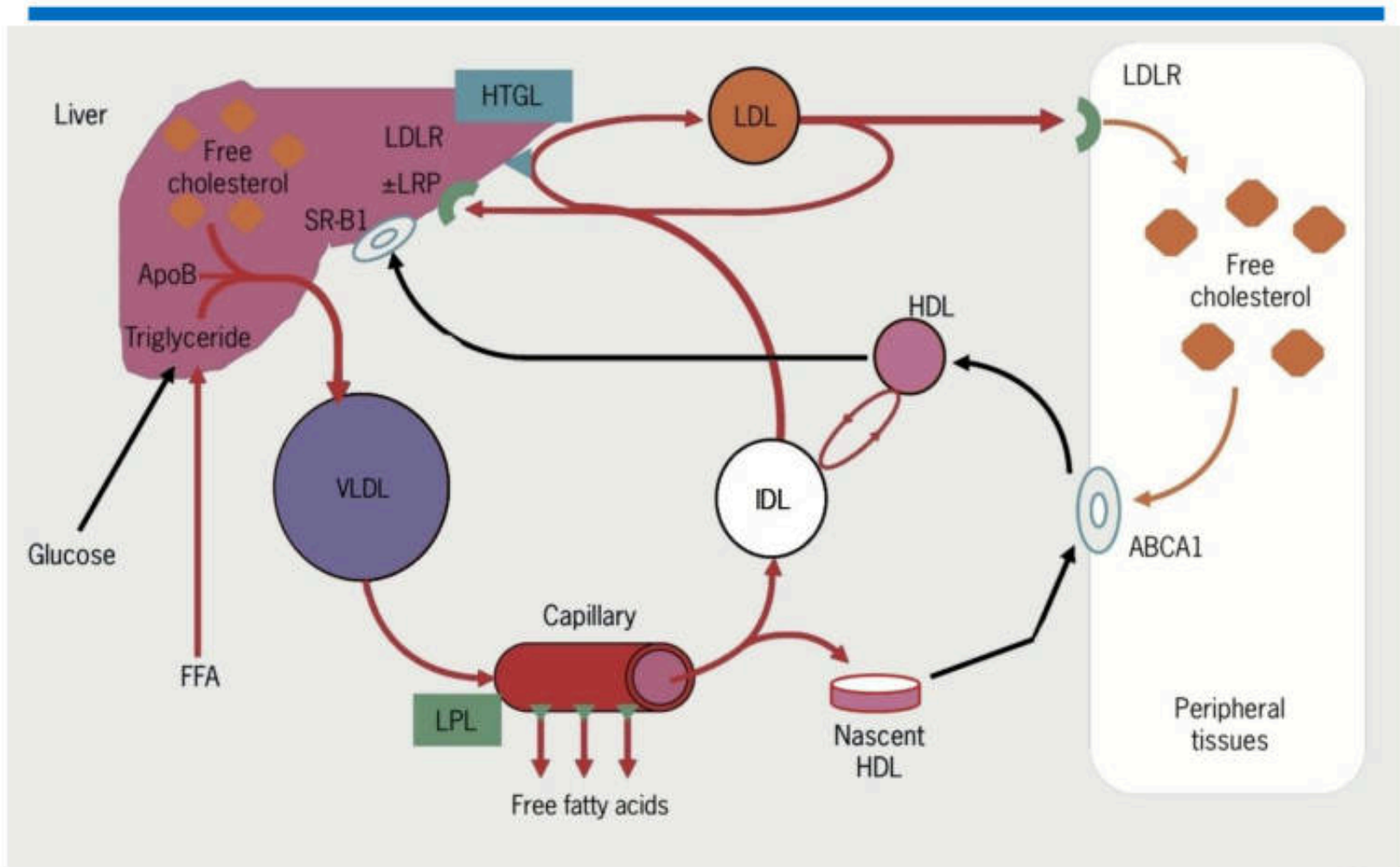
The endogenous lipoprotein pathway



The liver expresses a large complement of LDL receptors and removes about 75% of all LDL from the plasma

Key: ApoB = apolipoprotein B; FFA = free fatty acids; HDL = high-density lipoprotein cholesterol; HTGL = hepatic triglyceride lipase; IDL = intermediate-density lipoprotein cholesterol; LDLR = low-density lipoprotein receptor; LPL = lipoprotein lipase; LRP = LDL receptor-related protein; VLDL = very low-density lipoprotein cholesterol

HDL and the reverse cholesterol pathway



Key: ABCA1 = ATP-binding cassette, sub-family A; ApoB = apolipoprotein B; FFA = free fatty acids; HDL = high-density lipoprotein cholesterol; HTGL = hepatic triglyceride lipase; IDL = intermediate-density lipoprotein cholesterol; LDLR = low-density lipoprotein receptor; LPL = lipoprotein lipase; LRP = LDL receptor-related protein; SR-B1 = scavenger receptor class B member 1; VLDL = very low-density lipoprotein cholesterol



Reference values of plasma lipids in adults (>20 years)

Friedewald formula:

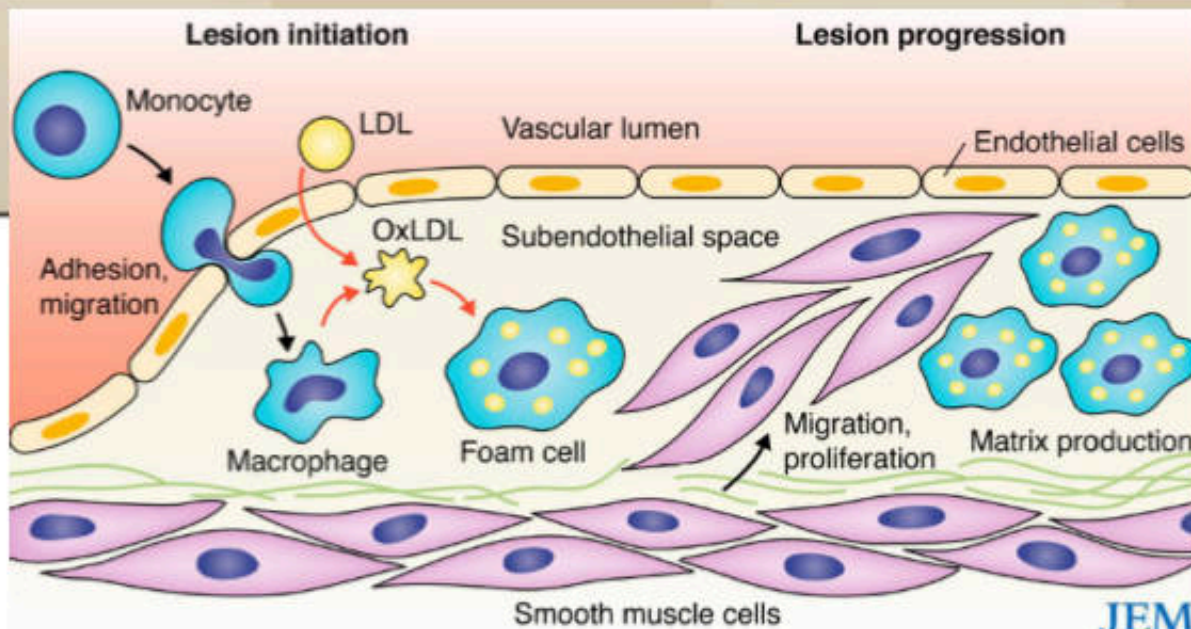
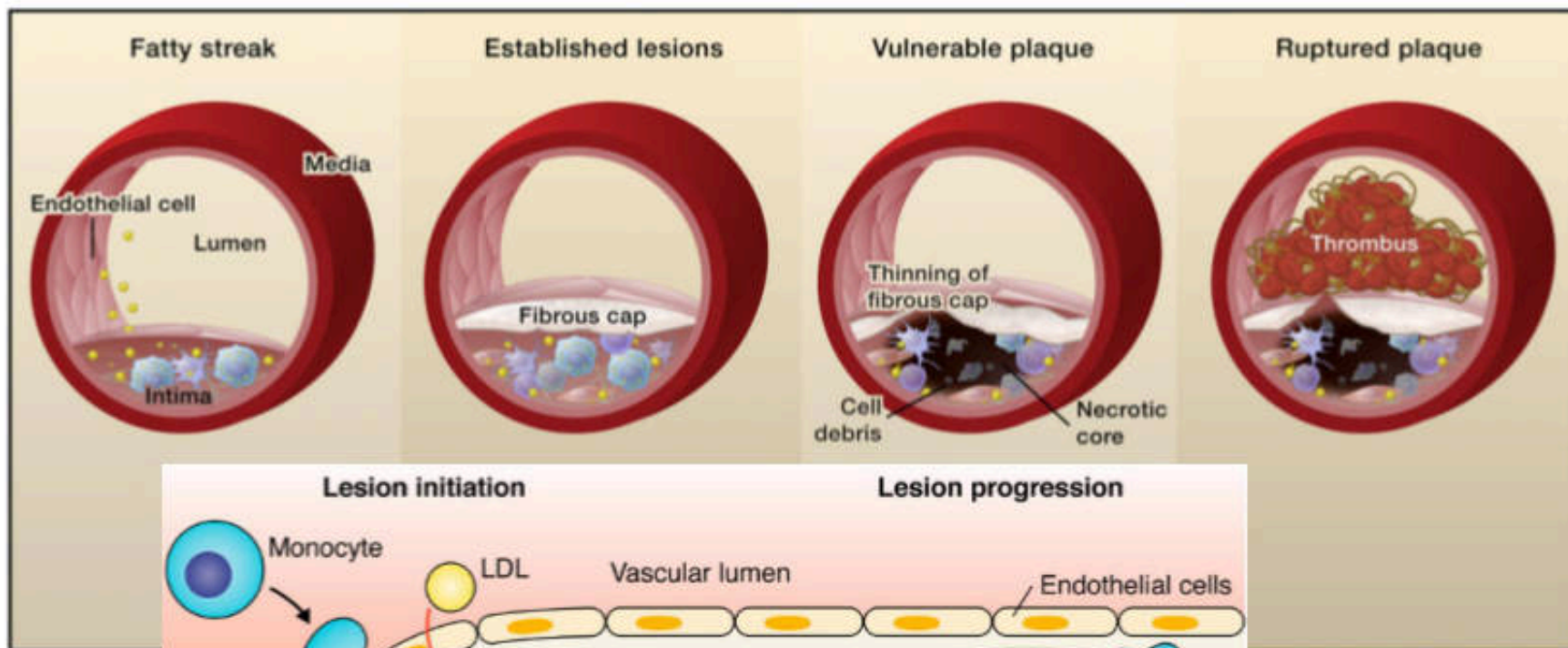
$$\text{LDL-C} = \text{TC} - \text{HDL-C} - (\text{TG}/2.2) \text{ in mmol/L}$$

or

$$\text{LDL-C} = \text{TC} - \text{HDL-C} - (\text{TG}/5) \text{ in mg/dL}$$

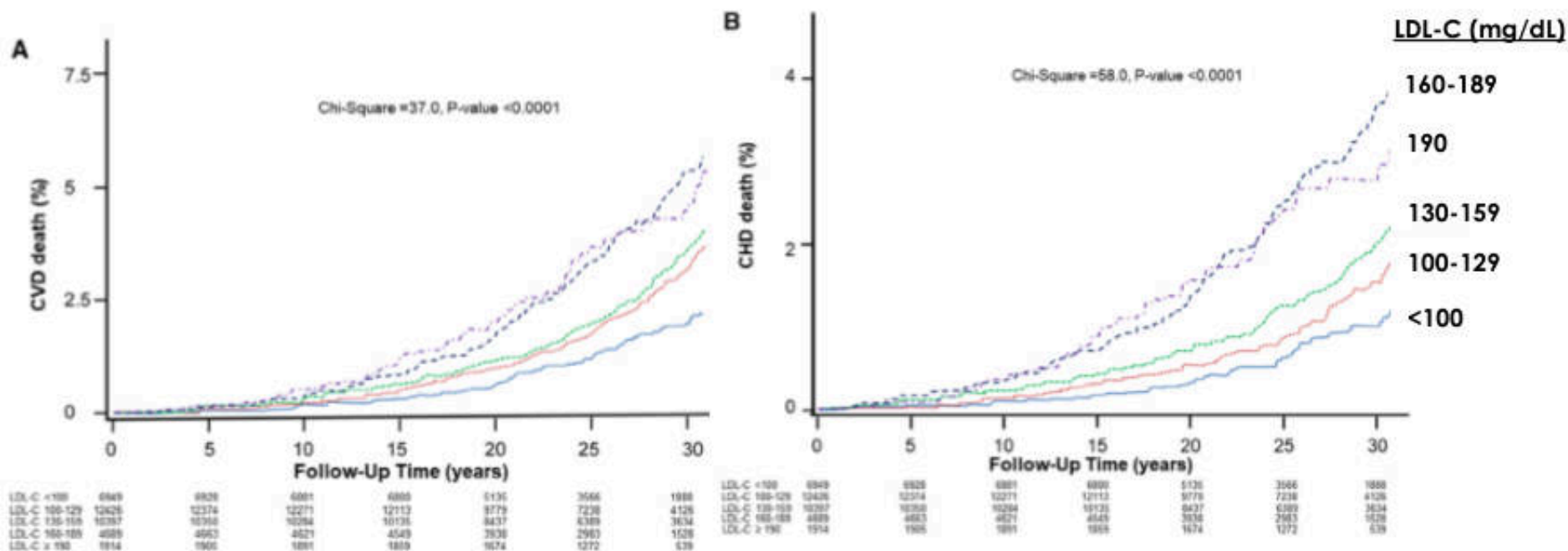
Lipids	Values (mg/dL)		
	Desirable	Borderline	Altered
TC	< 200	200-239	≥ 240
LDL-c	< 130	130-159	≥ 160
HDL-c	≥ 35	—	—
TG	< 200	—	≥ 200

LDL-C and the pathogenesis of atherosclerosis

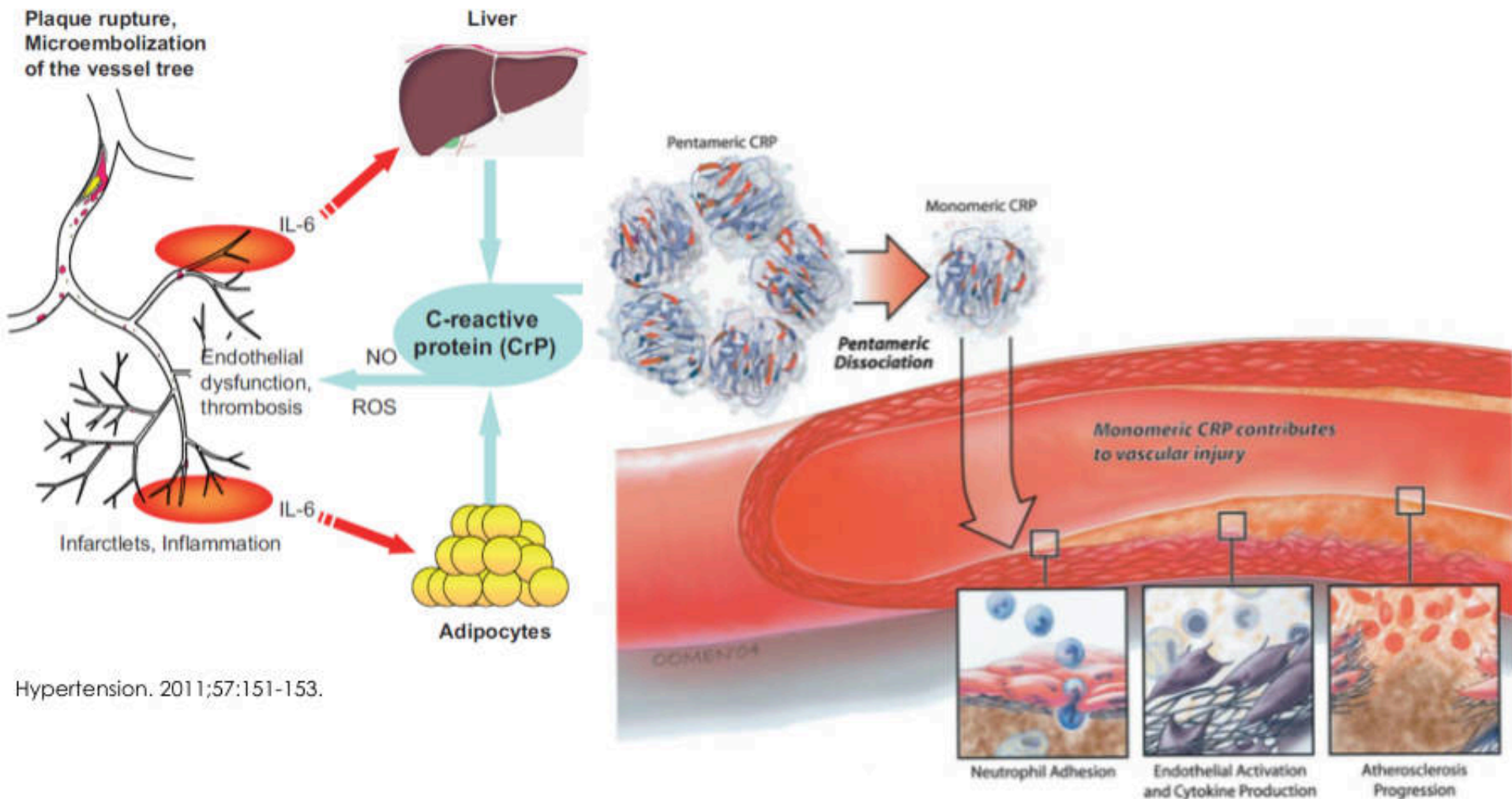




LDL-C and CVD and CHD mortality



CRP secretion in atherosclerosis

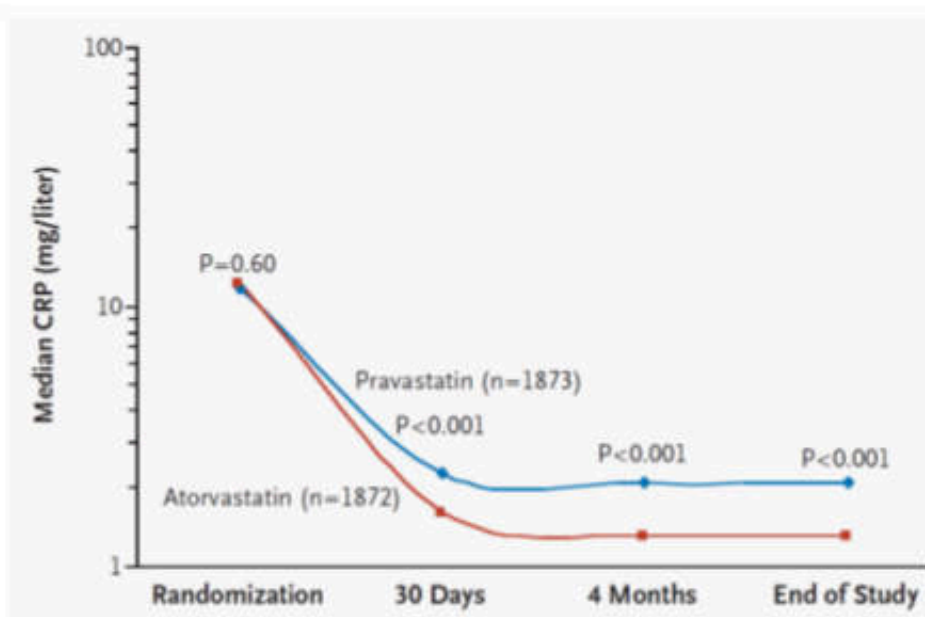
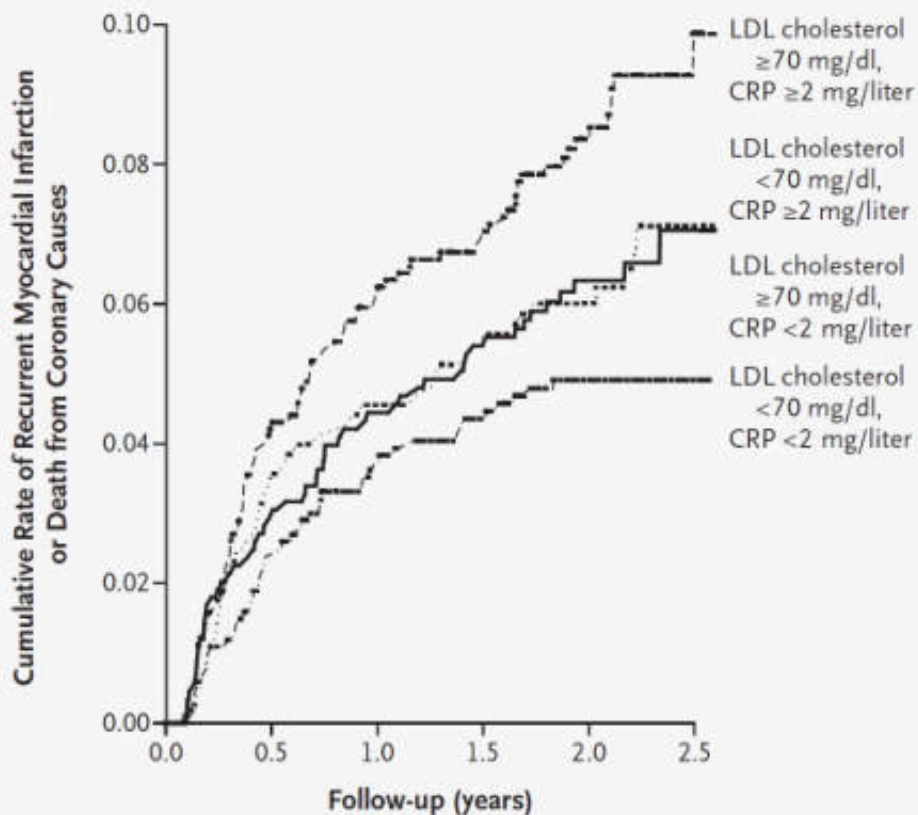


Hypertension. 2011;57:151-153.

Circulation. 2004;109:1914-1917.



Relationship between CRP and CVD events





Lipid-lowering drugs

1. HMG-CoA reductase inhibitors (Statins)
2. Adenosine triphosphate citrate lyase inhibitor
3. PCSK9 inhibitors
4. Inhibitor of cholesterol absorption
5. Bile acid sequestering agents

Primarily
reduce
LDL-C

6. Fibric acid derivatives (Fibrates)
7. Niacin (nicotinic acid)
8. Omega-3 fatty acid

Primarily
reduce TG



Lipid-lowering drugs (Cont.)

9. Microsomal triglyceride transfer protein (MTP) inhibitor

10. Inhibitor of apo B-100 synthesis

11. Inhibitor of angiopoietin-like protein 3

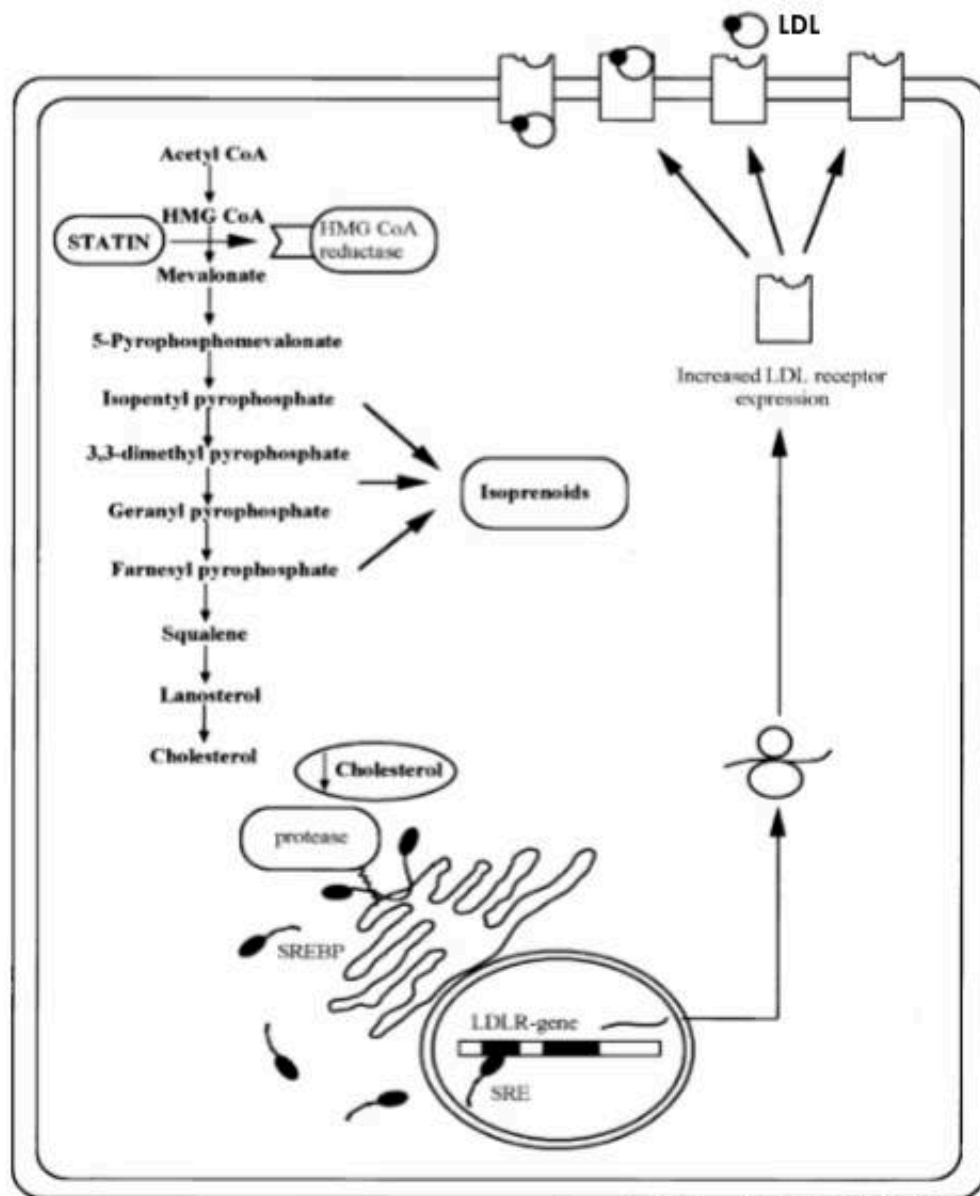
Primarily
used in FH

Self-study



HMG-CoA reductase inhibitors

- Mevalonic acid-like moiety
- **Competitively inhibits HMG-CoA reductase**
 - An early and **rate-limiting step** in **cholesterol biosynthesis**
- **Major effect: reduction of LDL**
- The reduction in hepatic VLDL production induced by statins is thought to be mediated by
 - Reduced synthesis of cholesterol, a required component of VLDLs





HMG-CoA reductase inhibitors

- Simvastatin and lovastatin (inactive lactones)
 - Must be transformed in the liver to their respective β -hydroxy acids (i.e. simvastatin acid, lovastatin acid)

Table 1. Relevant pharmacokinetic, pharmacodynamic and physiochemical properties of statins.

	Standard Daily Dose (mg)	Oral Bioavailability (%)	IC ₅₀ HMG-CoA Reductase (nM)	Aqueous Solubility (mg/mL)	Plasma Target Concentration (ng/mL) *	Elimination Half-Life (h)
Pravastatin	10–40 [43]	18 [43]	4 [43]	0.006 [59]	~1.0 ^a [60]	1.8 [4]
Lovastatin	10–40 [43]	5 [43]	2–4 [43]	0.0004 [59]	~1.7 ^a [39]	3 [4]
Simvastatin	10–40 [43]	<5 [43]	1–2 [43]	0.01 [59]	1.4 ^b [61]	2 [4]
Fluvastatin	80 [43]	30 [43]	3–10 [43]	0.0005 [59]	1.9 ^b [62]	1.2 [4]
Atorvastatin	10–80 [43]	12 [43]	1.16 [43]	0.000001 [59]	5.0 ^b [63]	14 [4]
Rosuvastatin	5–40 [43]	20 [43]	0.16 [43]	0.02 [59]	0.5 ^b [64]	19 [4]
Pitavastatin	1–4 [43]	60 [43]	0.1 [43]	4 × 10 ^{−4} [59]	2.5 ^b [65]	11 [4]

* Steady-state C_{trough} at 12 h ^a or 24 h ^b post dose.

Comparative efficacy of the different statins on various lipid fractions

Serum lipid	Rosuvastatin	Atorvastatin	Pitavastatin	Simvastatin	Pravastatin	Lovastatin	Cerivastatin ^a	Fluvastatin
↓LDL-C (%)	63	50	48	41	34	34	28	24
↑HDL-C (%) ^b	10	6	NA ^c	12	12	9	10	8
↓TG (%)	28	29	23	18	24	16	13	10

^aVoluntarily withdrawn from clinical use; ^bthis effect was elicited in patients with hypercholesterolaemia by a daily dose of 40 mg for atorvastatin, fluvastatin, lovastatin, pravastatin, simvastatin and rosuvastatin, 4 mg for pitavastatin and 0.3 mg for cerivastatin; ^cno significant effect reported. LDL-C = low-density lipoprotein cholesterol; HDL-C = high-density lipoprotein cholesterol; TG = triglyceride.

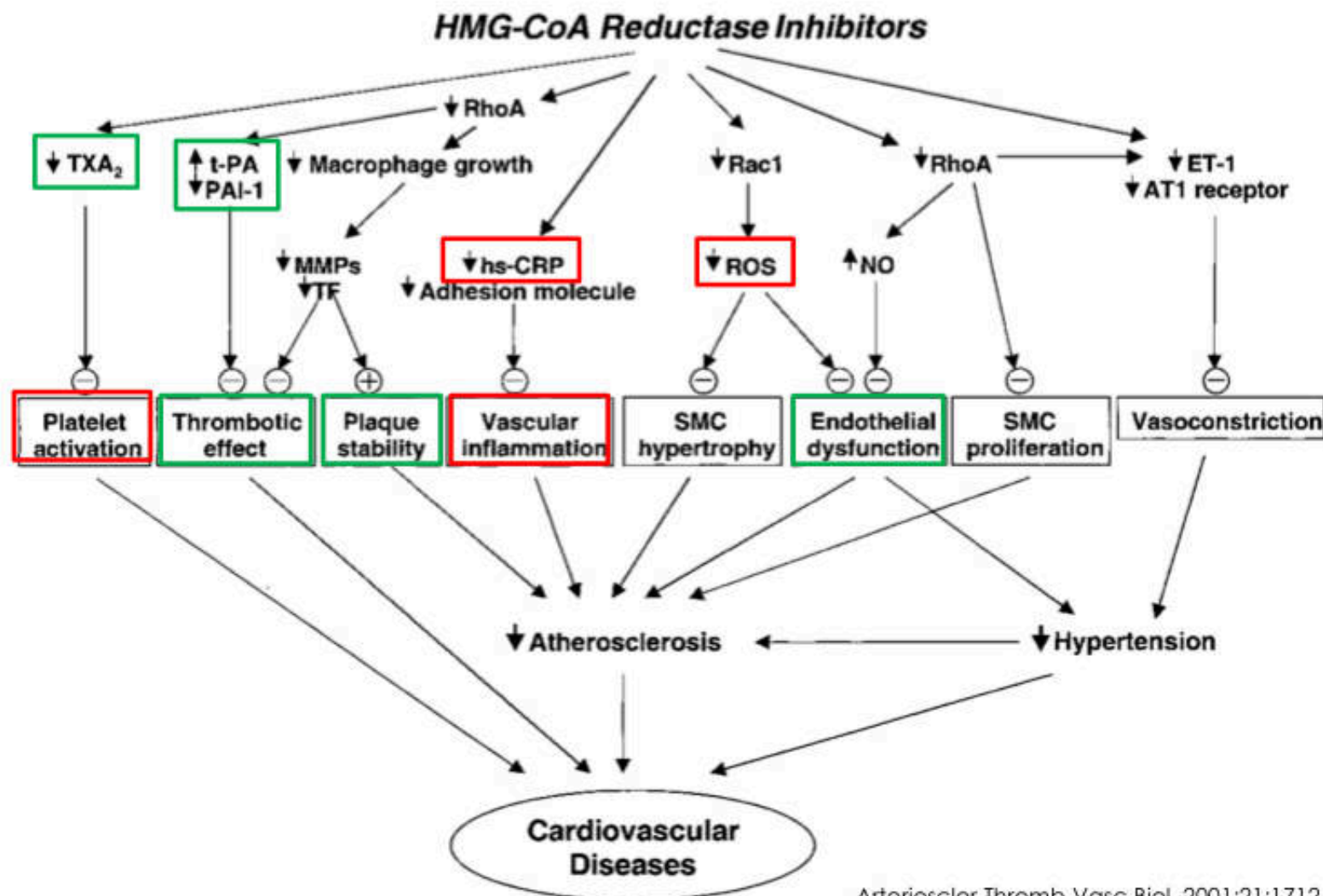
Fundam Clin Pharmacol. 2005;19(1):117-25.

High-Intensity (Lowers LDL-C ≥ 50%)	Moderate-Intensity (Lowers LDL-C 30-49%)	Low-Intensity (Lowers LDL-C <30%)
- Atorvastatin (40 mg[‡]) 80 mg - Rosuvastatin 20 (40 mg)	- Atorvastatin 10 mg (20 mg) - Rosuvastatin (5 mg) 10 mg - Simvastatin 20-40 mg[§] - Pravastatin 40 mg (80 mg) - Lovastatin 40 mg (80 mg) - Fluvastatin XL 80 mg - Fluvastatin 40 mg BID - Pitavastatin 1-4 mg	- Simvastatin 10 mg - Pravastatin 10-20 mg - Lovastatin 20 mg - Fluvastatin 20-40 mg

BID indicates twice daily; FDA, U.S. Food and Drug Administration; LDL-C, low-density lipoprotein cholesterol; RCT, randomized controlled trial; VOYAGER, an individual patient data meta-analysis Of statin therapy in At risk Groups: Effects of Rosuvastatin, atorvastatin and simvastatin; and XL, extended release



Pleiotropic effects of statins





PK of Statins

- Lovastatin absorption is increased when taken with food
- Extensive first-pass hepatic uptake of all statins, mediated primarily by the organic anion transporter OATP1B1
- The metabolites of all statins have some HMG-CoA reductase inhibitory activity
 - Except fluvastatin and pravastatin
- Pravastatin and its metabolites are only 50% protein-bound
- More than 70% of statin metabolites are excreted by the liver, with subsequent elimination in the feces
- The longer $t_{1/2}$ of atorvastatin and rosuvastatin may contribute to their greater cholesterol-lowering efficacy



Comparative pharmacological properties of statins

Parameters	Lovastatin	Simvastatin	Pravastatin	Fluvastatin	Atorvastatin	Rosuvastatin	Pitavastatin
Lactone prodrug	Yes	Yes	No	No	No	No	No
Bioavailability (%)	5	< 5	18	24 – 30	12	20	60 – 80
Solubility	Lipophilic	Lipophilic	Hydrophilic	Lipophilic	Lipophilic	Hydrophilic	Lipophilic
Protein binding (%)	> 98	> 95	50	> 98	> 98	88	> 99
T _{max} (h)	2.8	1.3 – 2.4	0.9 – 1.6	0.5 – 1.5	2 – 4	3 – 5	0.5 – 0.8
Half-life (h)	2 – 5	2 – 5	1 – 3	1 – 3	7 – 20	19	10 – 13
Hepatic extraction (%)	≥70	≥80	45	≥70	70	63	80
Renal excretion (%)	10	13	20	6	< 5	10	15
Metabolism	+++	+++	+	+++	+++	+	++
CYP	3A4/5, 2C8	3A4/5, 2C8	3A4 (Minimally)	2C9	3A4, 2C8	2C9, 2C19 (Minimally)	2C9, 2C8 (Minimally)
Influx transporters	SLCO1B1	SLCO1B1	SLCO1B1/2B1	SLCO1B1	SLCO1B1	SLCO1B1/1B3/ 2B1/1A2 SLC10A1	SLCO1B1/1B3 /2B1/1A2
Efflux transporters	ABCB1	ABCB1/G2	ABCB1/B11/ C2/G2	ABCG2	ABCB1/G2	ABCB1/C2/G2	ABCB1/C2/G2

Data taken from Refs. [11,14-16,93].

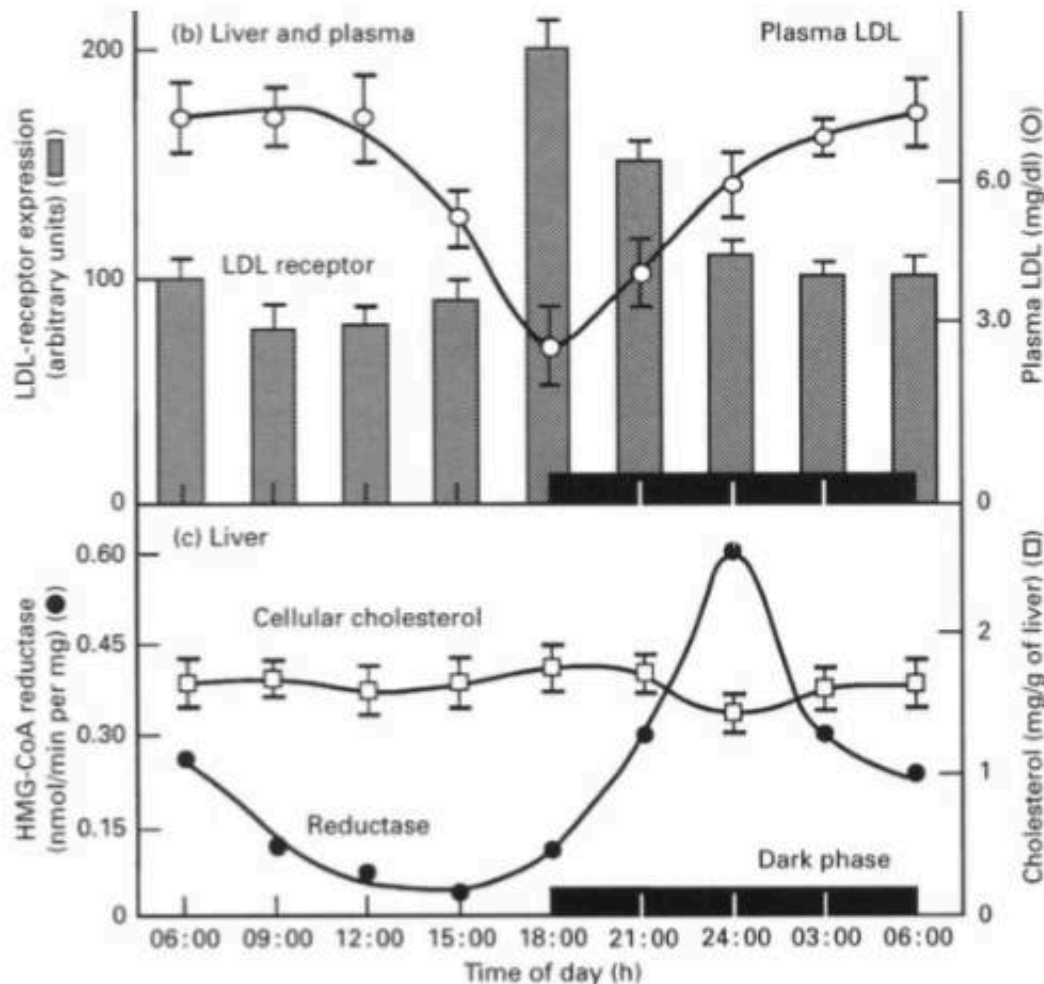
ABC: ATP-binding cassette; CYP: Cytochrome P450 enzymes; SLC: Solute carrier; SLCO: Solute carrier organic anion transporter.

SLCO1B1 = OATP1B1 = Organic anion transporting protein 1B1

Statin of choice for patients with severe renal dysfunction (no dose adjustment)

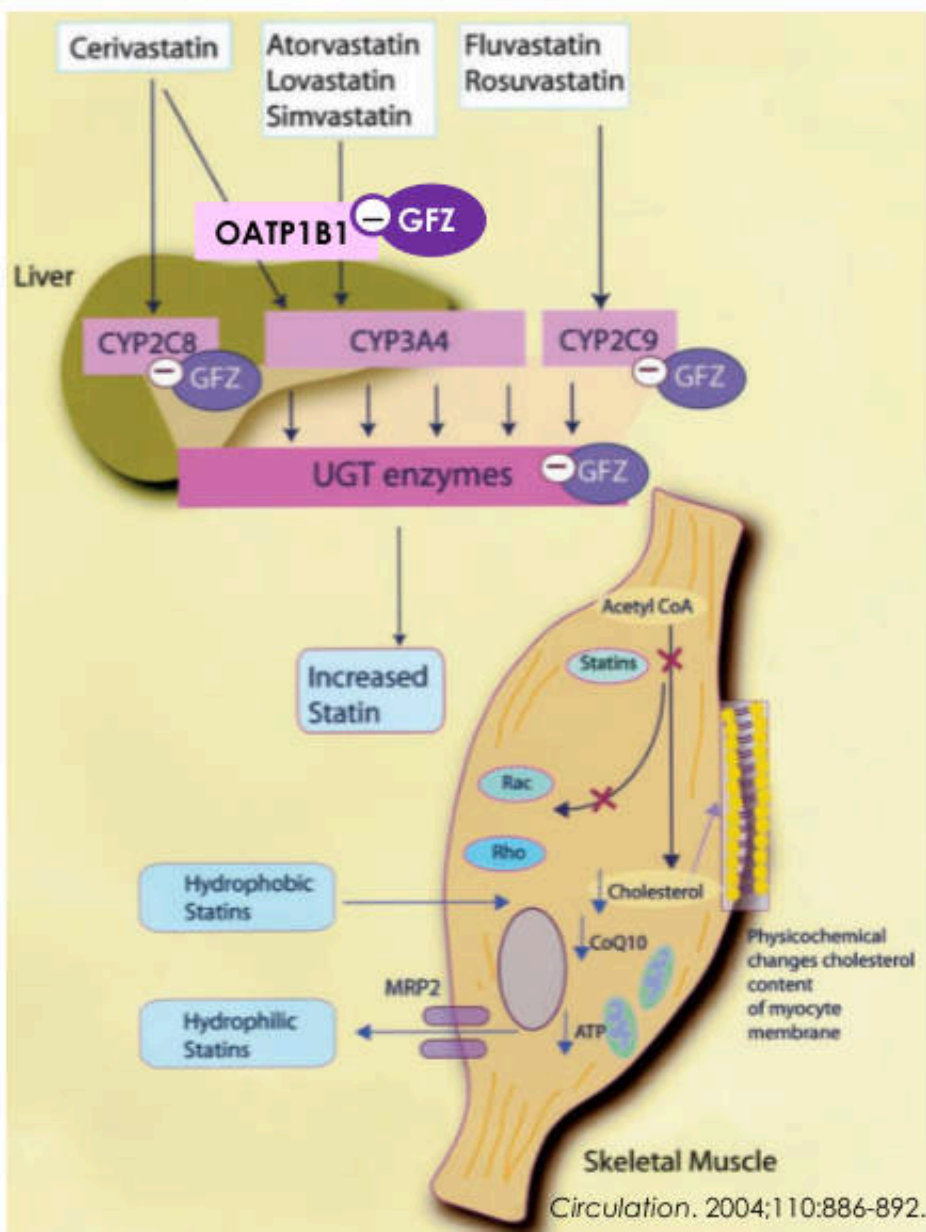


Diurnal pattern of cholesterol synthesis in liver



- Statins with $t_{1/2}$ of 4 h or less should be taken in the evening
- Atorvastatin and rosuvastatin both have longer $t_{1/2}$ and may be taken at other times of day to optimize adherence

ADRs of Statins



• Myopathy

Condition	Definition
Myopathy	General term to describe all skeletal muscle-related adverse effects
Asymptomatic CK elevation	CK elevation without muscle symptoms
Myalgia	Muscle pain or weakness without CK elevation
Myositis	Muscle symptoms with CK elevation typically $<10 \times \text{ULN}$
Rhabdomyolysis	Muscle symptoms with CK elevation typically $>10 \times \text{ULN}$, and with creatinine elevation (usually with brown urine and urinary myoglobin)

CK = creatine kinase; ULN = upper limit of normal.

Drug Saf 2010; 33 (3): 171-187

- Rosuvastatin and pravastatin may be better tolerated than others
 - Should be considered in patients with a history of myalgias with other statins

• Hepatotoxicity (very rare)



Drug interaction with statins

Non-hypolipidaemic medicines (CYP3A4 inhibitors)

Ciclosporin

Macrolide antibacterials (erythromycin, clarithromycin)

Azole antifungals (itraconazole, ketoconazole, fluconazole)

Calcium channel antagonists (diltiazem, verapamil)

Nefazodone

HIV protease inhibitors (ritonavir, nelfinavir, indinavir)

Warfarin

Histamine H₂ receptor antagonists (cimetidine, ranitidine)

Omeprazole

Amiodarone

Hypolipidaemic medicines

Fibrates (gemfibrozil > bezafibrate, clofibrate, fenofibrate)

Niacin

Other substances

Grapefruit juice (CYP3A4 inhibitor)

Over-the-counter medications (Chinese red rice fungus)

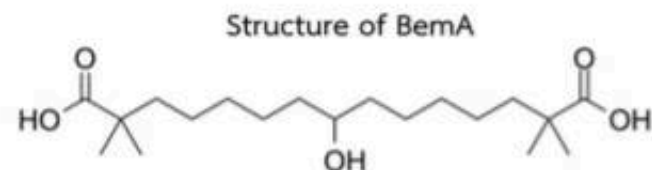
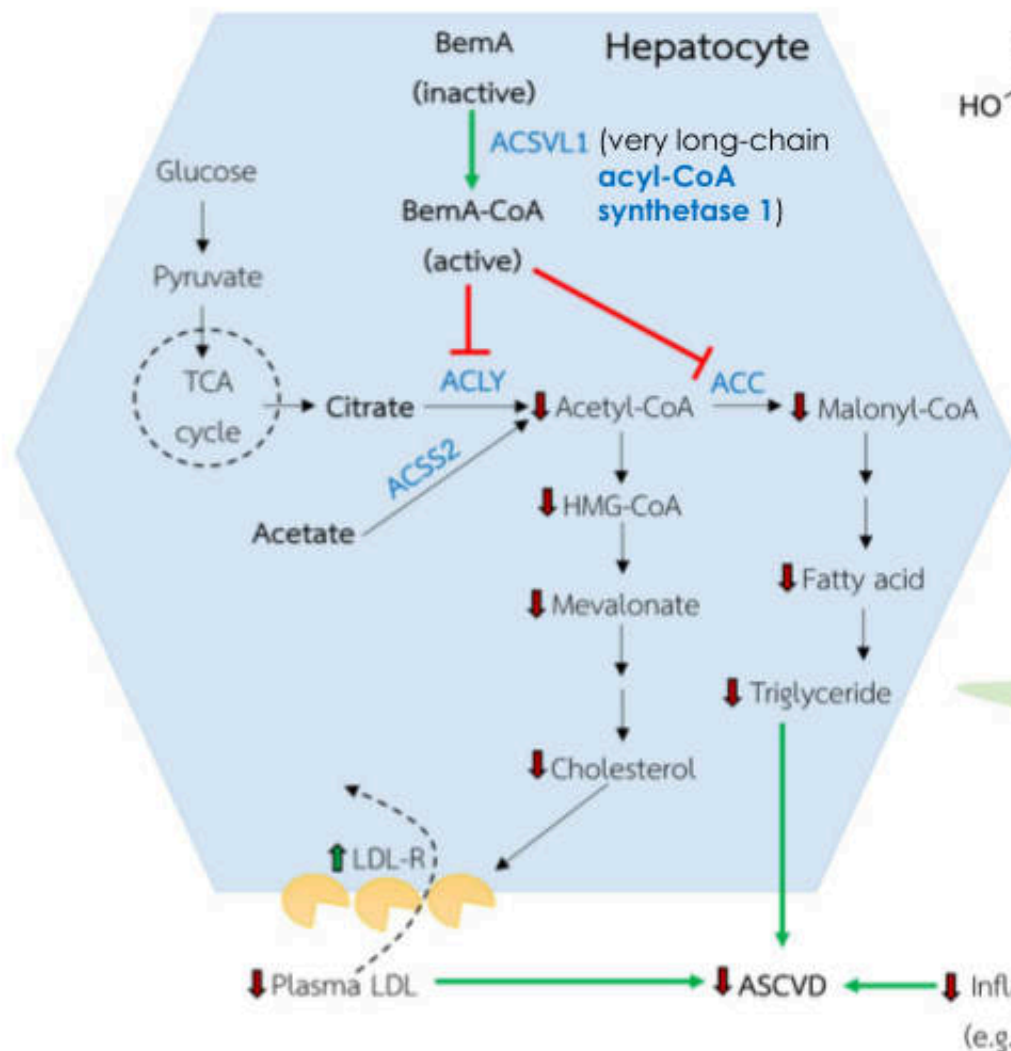
May precipitate
statin-induced
myopathy

Pravastatin, pitavastatin, fluvastatin, and rosuvastatin are not extensively metabolized by CYP3A4,

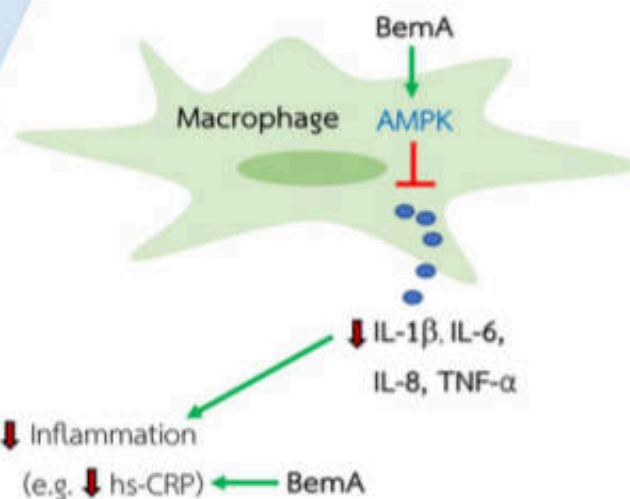
- These statins may be less likely to cause myopathy when used with one of the predisposing drugs



Bempedoic acid: Inhibition of adenosine triphosphate-citrate lyase



(ACLY = Adenosine triphosphate-citrate lyase)



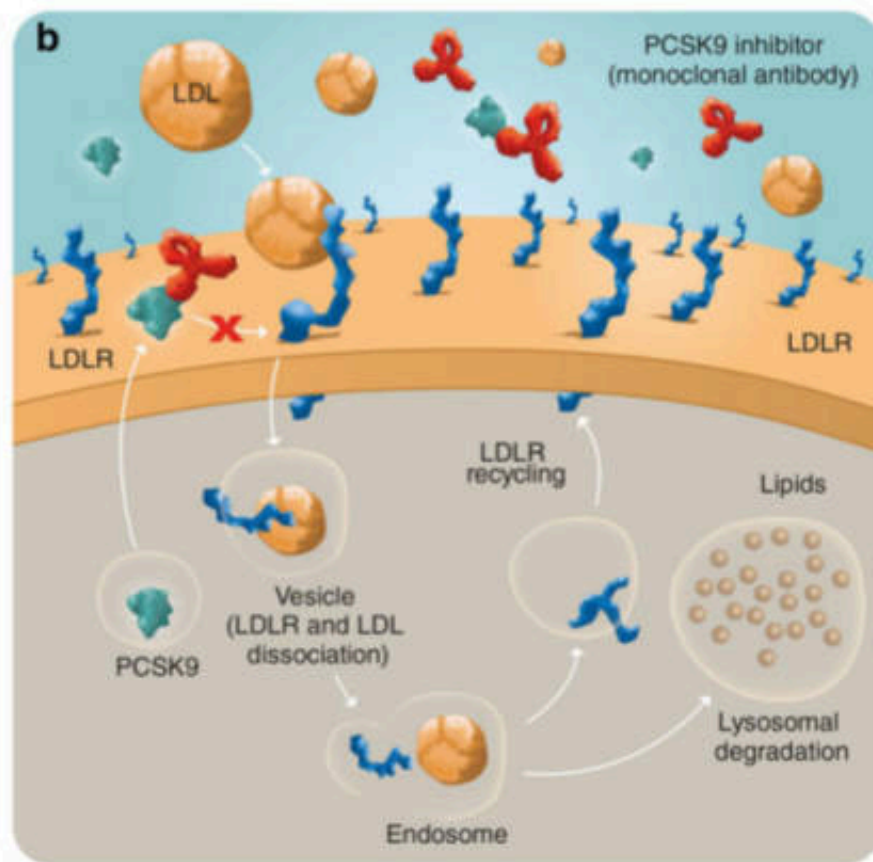
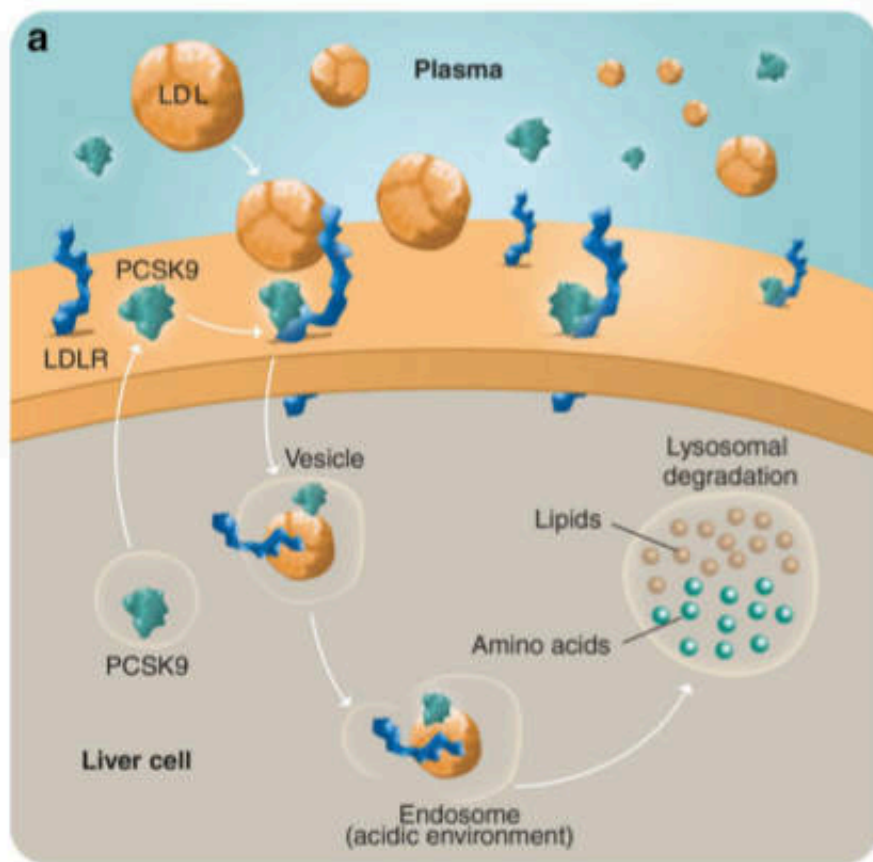


Bempedoic acid

- **Reduce hepatic cholesterol biosynthesis**
 - Increase in LDL receptor activity
 - Reduction in the plasma LDL-C (25-30%)
- **Reduce hs-CRP (~20%)**
- **Reduce ASCVD risk in statin-intolerant patients** (CLEAR-OUTCOMES trial)
- Orally active
- Half-life (15-24 hrs)
- **ADRs**
 - Most common: Headache (10-20%)
 - Others: constipation (7%), myalgia (1-7%)
 - Since bempedoic acid requires hepatic conversion to active form, it is expected to cause low/no muscle toxicity relative to statins

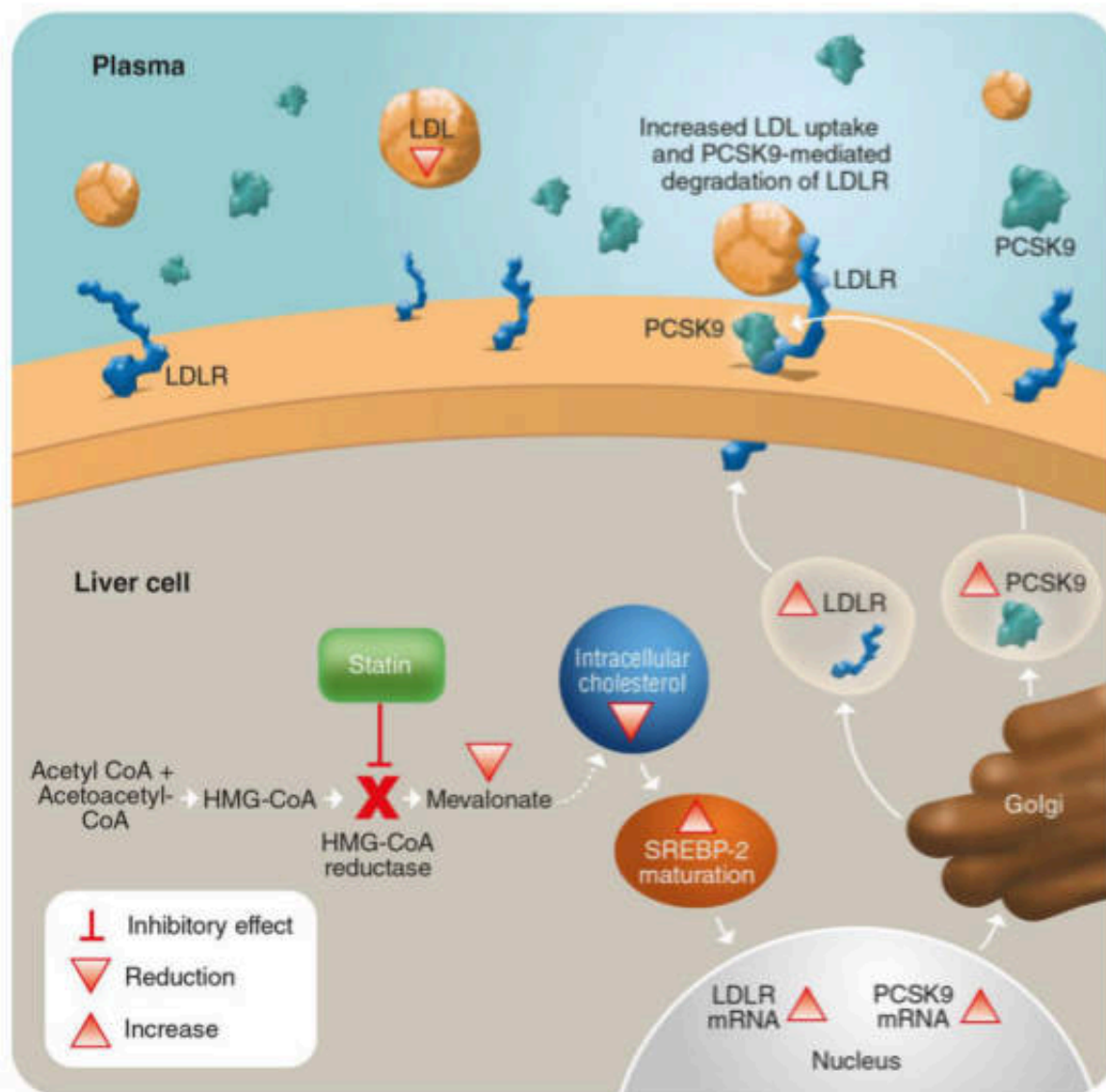
PCSK9 inhibitors

- Proprotein convertase subtilisin/kexin type 9 is a protease that binds to the LDL receptor on the surface of hepatocytes and enhances lysosomal degradation of the LDL receptor





Effect of statins on PCSK9 expression

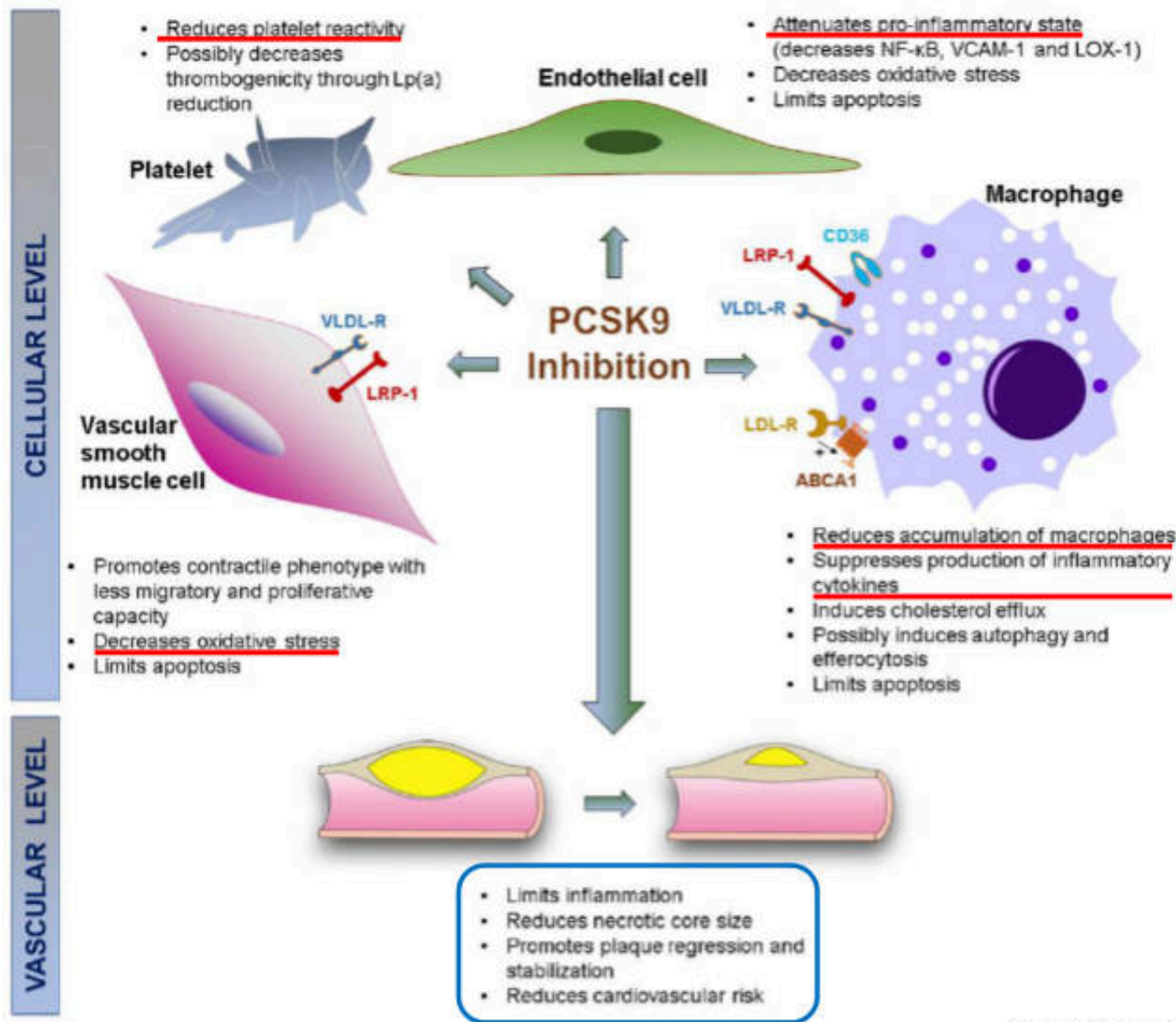


SREBP-2 = sterol-regulatory element-binding protein-2

Comparative pharmacological properties of PCSK9 inhibitors

	Alirocumab	Evolocumab
Pharmacodynamics		
• Time of maximum suppression of free PCSK9 (h)	4-8	4 (Max. suppression 85-95%)
• Time to nadir of LDL-C (days)	15	• 14 (140 mg) • 21 (420 mg)
• Reduction on LDL-C (%)	58 (150 mg)	64 (420 mg) equivalent to statins
Pharmacokinetics		
• Bioavailability (%)	85	72
• Tmax (days)	3-7 (median)	3-4 (median)
• Time to steady state (weeks)	4-6	12
• Half-life (days)	• 17-20 • 12 (with statin co-administration)	11-17
• Route of administration	• 75-150 mg SC Q2W	• 140 mg SC Q2W • 420 mg SC QM

Pleiotropic effects of PCSK9 inhibitors





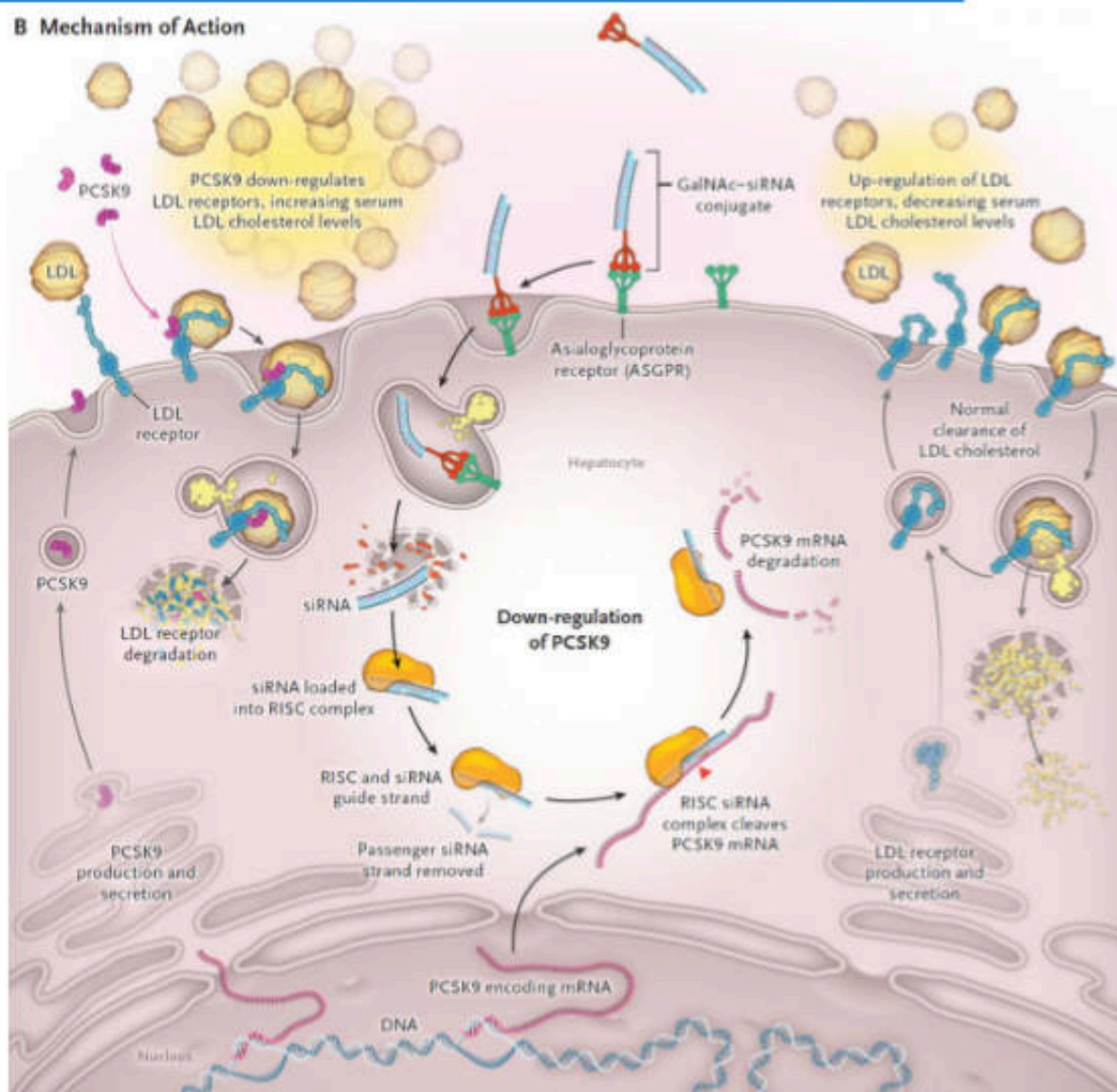
ADRs and drug interactions of PCSK9 inhibitors

- Injection site reactions are the most frequent adverse effect (<10% of patients)
- Slightly increase risk of infections (similar to other monoclonal antibodies)
 - Nasopharyngitis
 - Urinary tract infections
 - Upper respiratory infections
- Small (<1%) risk of neurocognitive effects
- No expected drug interactions with PCSK9 inhibitors

Inclisiran

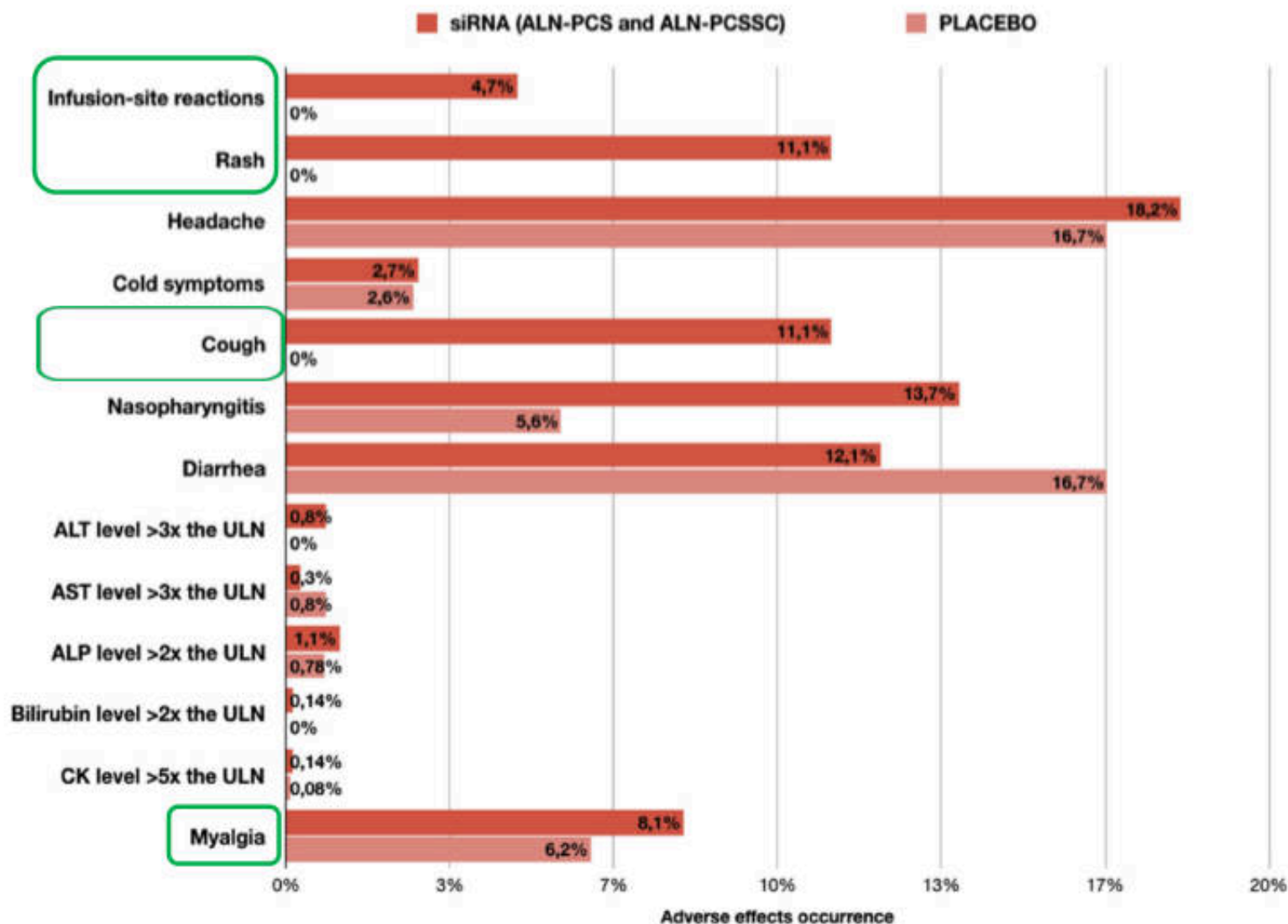
- A small interfering RNA (siRNA)
- First injection (day 1) s.c. and the 2nd dose at 3 months, followed by every 6-month injections
- **Plasma half-life:** 9.58 h
- **Tissue half-life**
 - At injection site: 526 h
 - Liver: 270 h
 - Kidney: 360 h

B Mechanism of Action



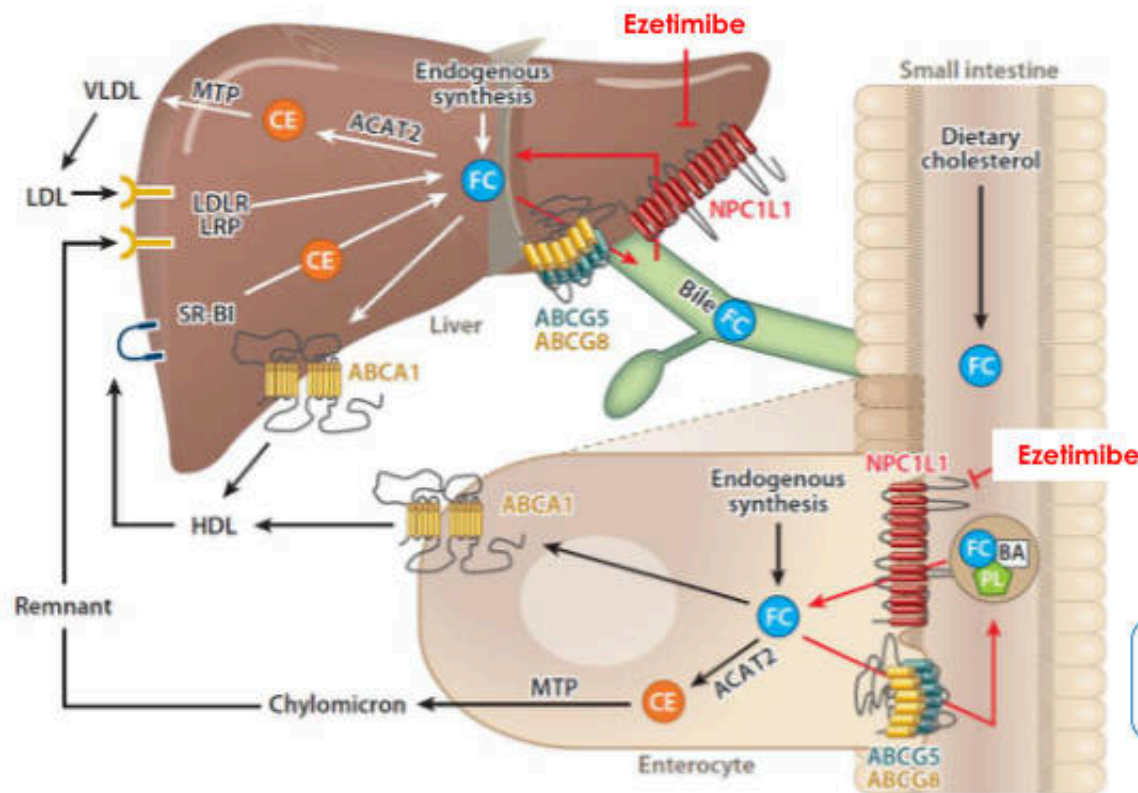


ADRs of Inclisiran



Ezetimibe

Active form: Ezetimibe glucuronide



NPC1L1 = Niemann-Pick C1-Like 1

FC = free cholesterol

CE = Cholesteryl ester

ABCG = ATP-binding cassette transporter

↓ Delivery of intestinal cholesterol to the liver by chylomicron remnants

Stimulates expression of the hepatic genes

↑ Hepatic LDL receptor

↑ Cholesterol biosynthesis

↑ LDL-C clearance from the plasma

Ezetimibe reduces LDL-C levels by 15%–20%.



Possible pleiotropic effects of ezetimibe

Parameter	Drug action	Remarks
LDL-C	Reduction by 7 to 18% when used as monotherapy, 13.8 to 25.1% on top of a statin	Great inter-individual variability
HDL-C	Increase by 1.3 to 6.2%	Not always significant
Triglycerides	Decrease by 1.7–9.4%	Perhaps greater in individuals with higher levels before treatment
hsCRP	Decrease by 10% on top of statin	Not significant reduction when used as monotherapy
Atherosclerosis in animal models	Overall, substantial atheroprotective effect	Possible improvement of liver status
Small dense LDL particles	Possible reduction of sdLDL	Especially in individuals with high levels before treatment
Endothelial function	Possible beneficial effect	Especially in individuals with insulin resistance

sdLDL: small dense LDL particles, hsCRP: high sensitivity CRP.



Ezetimibe

- Dual therapy with statins prevents both the enhanced cholesterol synthesis (induced by ezetimibe) and the increase in cholesterol absorption (induced by statins), providing additive reductions in LDL-C levels
- **PKs**
 - Ezetimibe is highly water insoluble
 - After ingestion, it is glucuronidated in the intestinal epithelium and absorbed and then enters an enterohepatic recirculation
 - Bile acid sequestrants inhibit its absorption
 - About 70% is excreted in the feces and about 10% in the urine (as a glucuronide conjugate)
 - No dose adjustment in mild hepatic dysfunction and all grade of renal insufficiency



Bile acid sequestering agents

- **Cholestyramine, Colestipol, Colesevelam**

- Highly positively charged molecules that are not absorbed from the intestine

Bind negatively charged bile acids, then excreted in the stool

Because >95% of bile acids are reabsorbed (enterohepatic circulation), interruption of this process ↓ the pool of bile acids

↑ Hepatic bile acid synthesis

↓ Hepatic cholesterol content
↑ LDL receptors (similar to statins)

↓ LDL-C levels (10-20%)

↑ Cholesterol synthesis caused by upregulation of HMG-CoA reductase

↑ Triglyceride synthesis



Adverse effects and drug interactions of Bile acid sequesterants

- **ADRs**

- Drinking a slurry of powdered cholestyramine or colestipol produces a gritty sensation
- GI side effects: Bloating, dyspepsia, constipation
- Hypertriglyceridemia

- **Drug interactions**

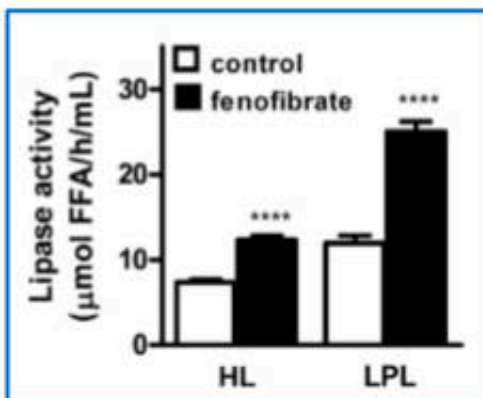
- Reduces absorption of other drugs
- Taken 1 h before or 3-4 h after other drugs

Fibrates

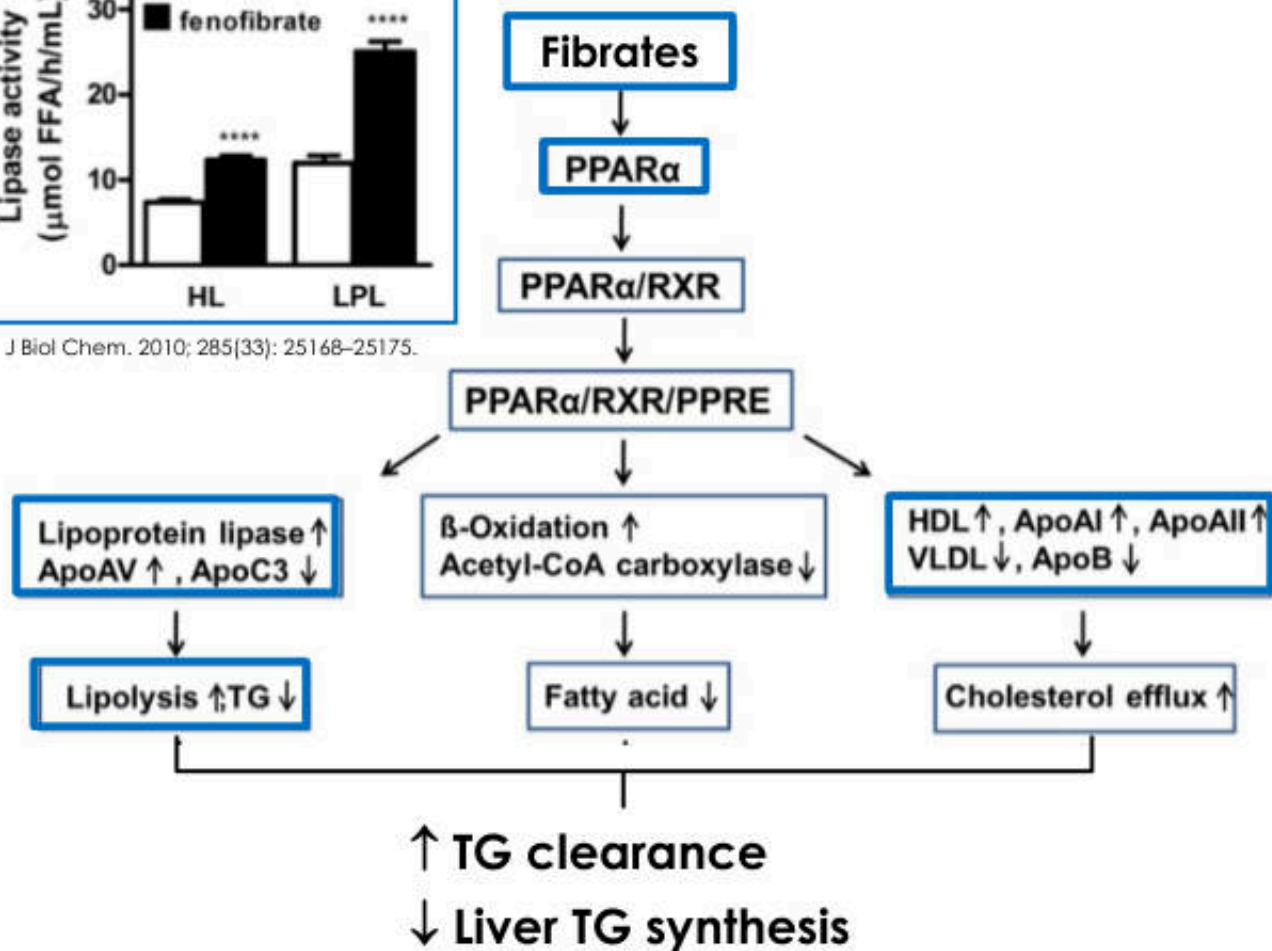
- **Gemfibrozil**
- **Fenofibrate**
- **Bezafibrate**
- Clofibrate

- Fenofibrate is more effective than gemfibrozil at increasing HDL levels

- Most fibrates have potential antithrombotic effects
 - Inhibition of coagulation
 - Enhancement of fibrinolysis



J Biol Chem. 2010; 285(33): 25168–25175.





Fibrates

- **The drugs of choice for treating severe hypertriglyceridemia (TG > 1000 mg/dL) and the chylomicronemia syndrome**
 - **↓ TG prevents episodes of pancreatitis**
- Patients with type III hyperlipoproteinemia (*dysbetalipoproteinemia*)
 - Are among the most sensitive responders to fibrates
 - Elevated TG and cholesterol levels are dramatically lowered
- In patients with mild hypertriglyceridemia (TG < 400 mg/dL)
 - **Fibrate treatment ↓ TG levels by up to 50%**
 - **↑ HDL-C concentrations by about 15%**



PKs of fibrates

- Are absorbed rapidly and efficiently (>90%) when given with a meal
- $T_{\max}$ 1–4 h
- More than 95% of these drugs in plasma are bound to protein, nearly exclusively to albumin
- The $t_{1/2}$: 1–2 h (gemfibrozil, bezafibrate), **20 h (fenofibrate)**
- The fibrates are excreted predominantly as glucuronide conjugates (60%–90%) in the urine
 - **Excretion is impaired in renal failure**



Adverse effects and drug interactions of fibrates

- **ADRs**

- Common: GIs (<5%)
- Myalgias
- Minor increase in liver transaminases and alkaline phosphatase
- Risk of gallstone

- **Drug interactions**

- ↑ Warfarin effect
- Gemfibrozil and statins
 - ↑ Statin level (PK interaction)
 - ↑ Myopathy (PD interaction)

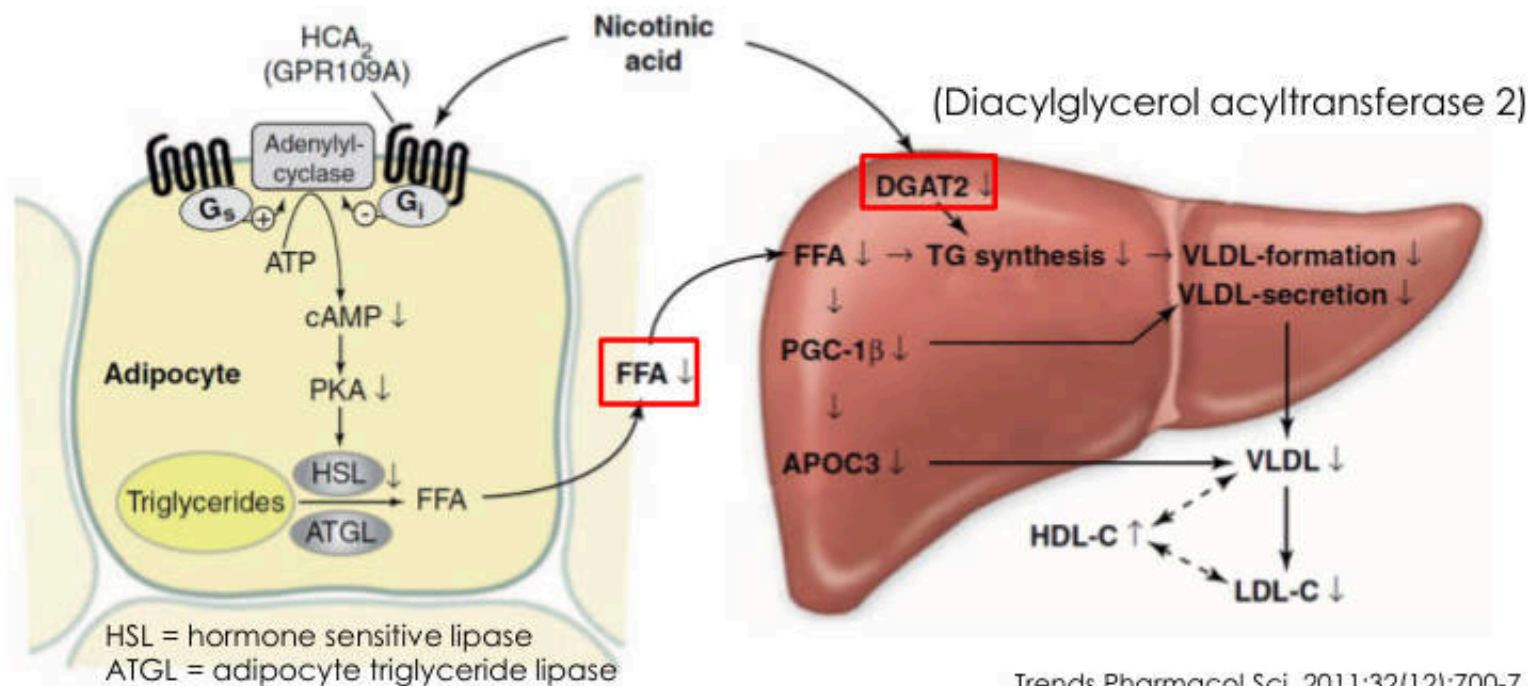


Niacin

- A water-soluble B-complex vitamin that functions as a vitamin only after conversion to nicotinamide derivatives (NAD or NADP)
- Only niacin itself affects lipid levels
 - Require larger doses than are required for its vitamin effects
- **PKs**
 - The doses of regular (crystalline) niacin are almost completely absorbed
 - $C_{\max}$ are achieved within 30–60 min
 - The $t_{1/2}$ is ~ 60 min, which necessitates dosing BID or TID
 - At higher doses, a greater proportion is excreted in the urine as unchanged

Niacin

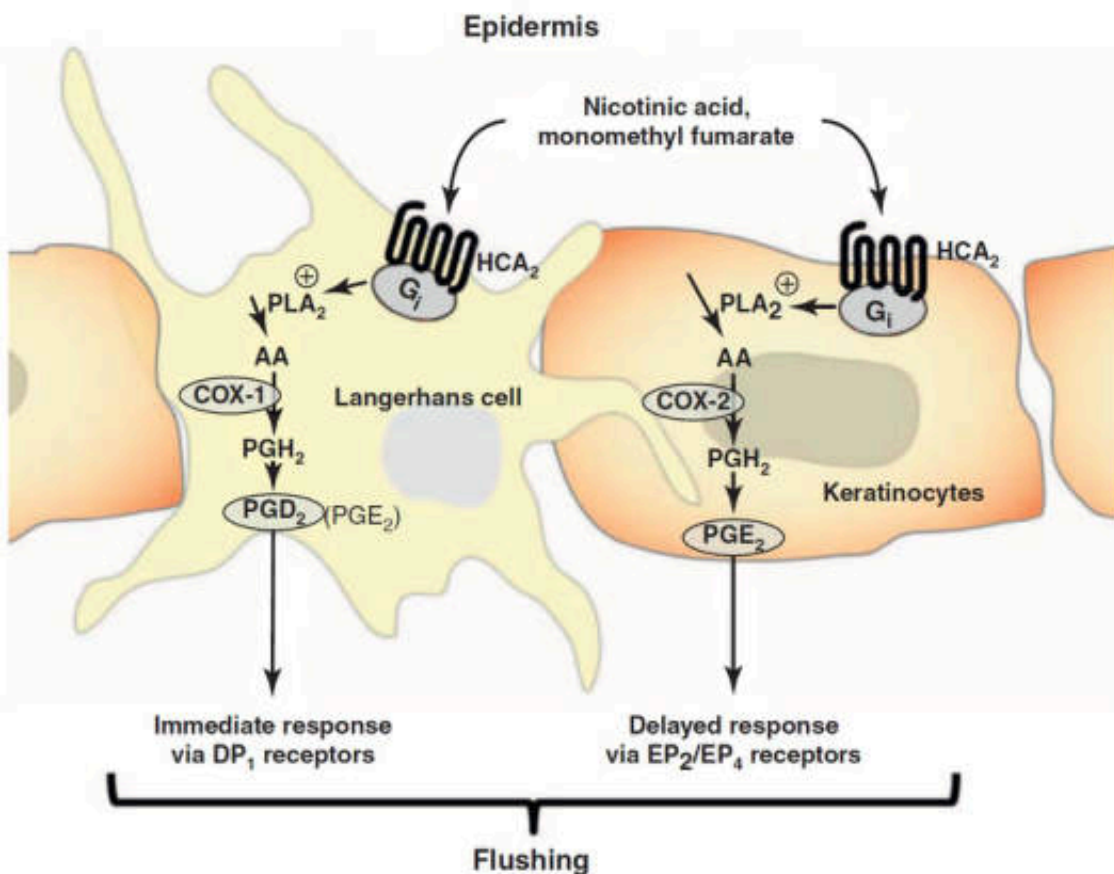
Hydroxy-carboxylic acid receptor 2



Trends Pharmacol Sci. 2011;32(12):700-7.

- Indicated for hypertriglyceridemia and elevated LDL-C
- Crystalline niacin in doses of 2–6 g/d **reduces triglycerides by 35%–50%** (as effectively as fibrates and statins)
- **The most effective agent available for increasing HDL-C (30%–40%)**
- Reductions of 25% in LDL-C levels are possible in high dose

ADRs of Niacin



- **Most common**
 - **Flushing** and skin reactions
 - **Dyspepsia**, nausea, vomiting, diarrhea
- Hepatotoxicity
 - ↑ AST, ALT
 - ↓ Serum albumin
- Myopathy (when used with statins)
- Hyperglycemia
- Hyperuricemia

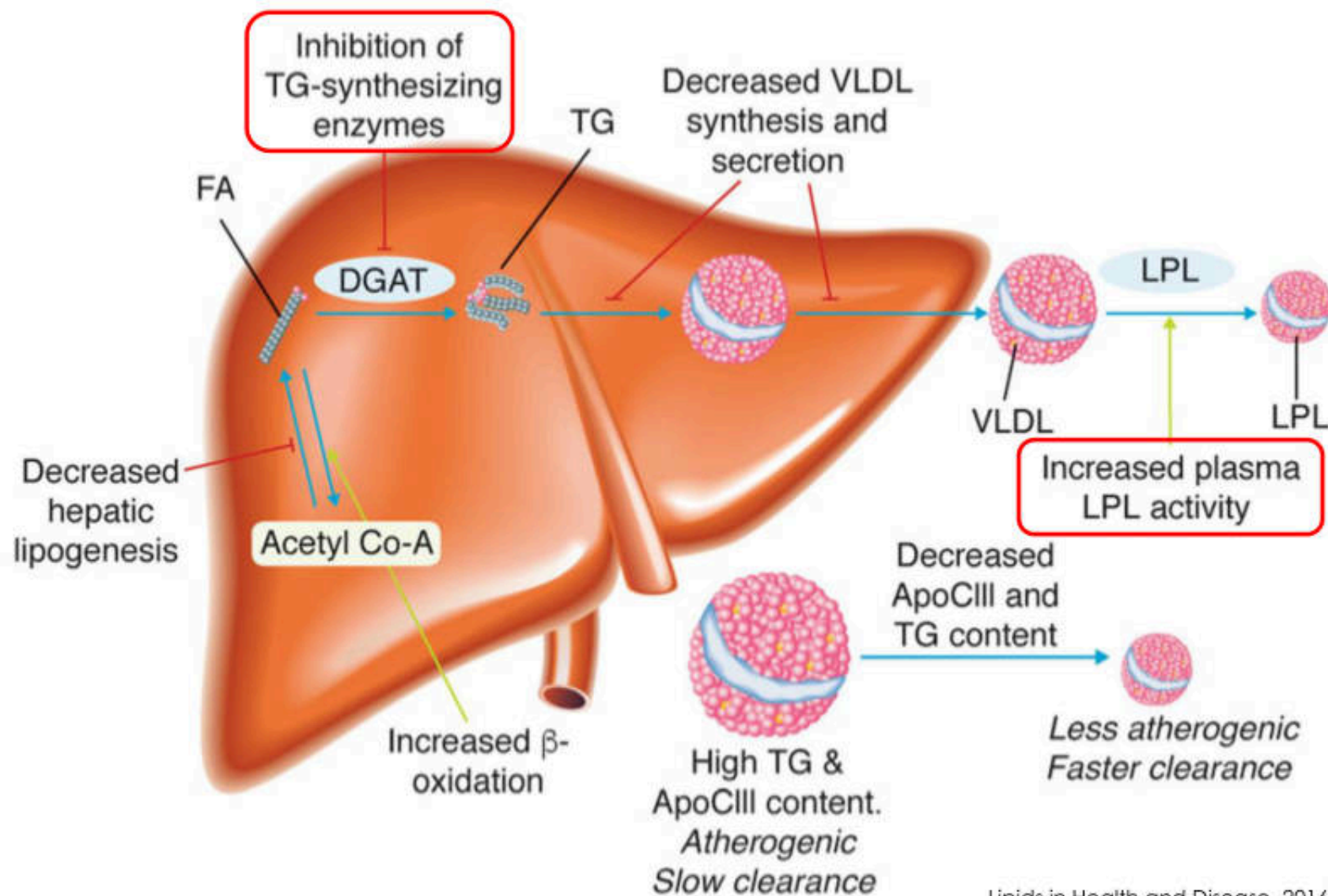


Omega-3 fatty acid

- Omega-3 fatty acids, commonly EPA and DHA ethyl esters, **reduce VLDL triglycerides**
 - FDA indications for triglycerides ≥ 500 mg/dL
- **PKs**
 - Fish oil 3–4 g/d administered with food
 - The small intestine absorbs EPA and DHA
 - Mainly oxidized in the liver, similar to fatty acids derived from dietary sources
 - The $t_{1/2}$ of elimination is approximately 50 to 80 h



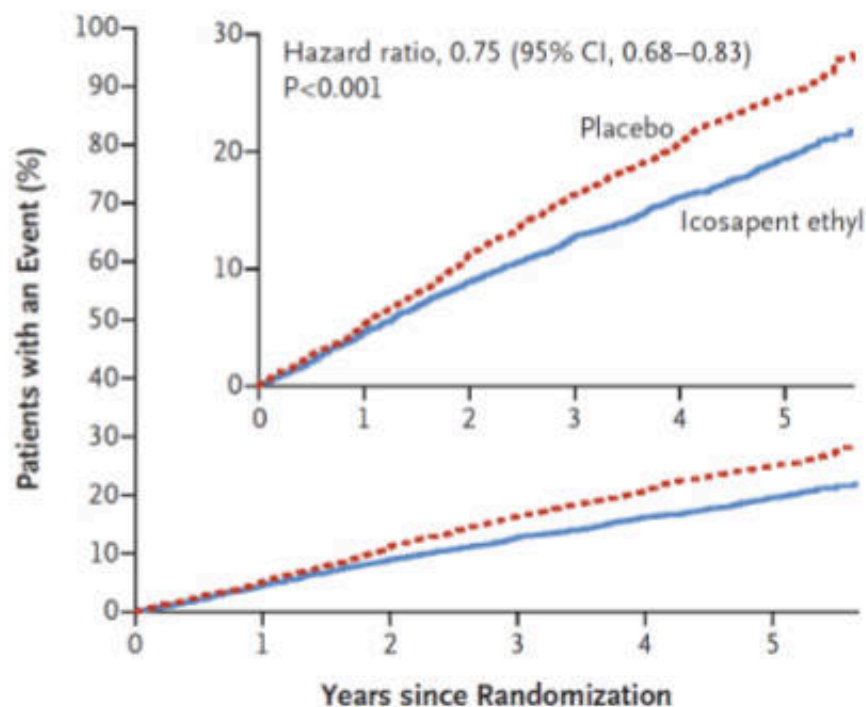
Proposed mechanisms of action of omega-3 fatty acid





Cardiovascular Risk Reduction with Icosapent Ethyl for Hypertriglyceridemia (REDUCE-IT)

Fasting triglyceride level of 135 to 499 mg/dl, despite the use of statins



No. at Risk

Placebo	4090	3743	3327	2807	2347	1358
Icosapent ethyl	4089	3787	3431	2951	2503	1430

- Icosapent Ethyl
 - An ethyl ester derivative of EPA (does not contain DHA)
- 2 g of icosapent ethyl twice daily
 - ↓ TG ~20%
 - ↓ hs-CRP ~40%
 - ↓ Risk of ischemic events, including cardiovascular death (25-30%)



ADRs

- Nausea, fishy burps, dyspepsia
- Increased LDL
- Arthralgia
- Because omega-3 fatty acids may prolong bleeding time, patients taking anticoagulants should be monitored

Thank You!

